

Qatar-1b

R-0628

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_230605 · created 2026-07-25T14:32:21+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460099.4119 +/- 0.0026 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.191 +/- 0.056
Transit depth (Rp/Rs)^2	3.67 +/- 2.16 [%]
Orbital Inclination (inc)	86.82 +/- 2.86
Airmass coefficient 1 (a1)	2.53 +/- 1.7
Airmass coefficient 2 (a2)	-0.6 +/- 1.41
Scatter in the residuals of the lightcurve fit is	71.28 %
Best Comparison Star	#1 - [177, 308]
Optimal Aperture	0.99
Optimal Annulus	5.75
Transit Duration (day)	0.0214 +/- 0.0057

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.191 ± 0.056 vs 0.1463 (deviation 0.61σ)
Duration fit vs published	0.0214 d vs 0.0692 d (deviation 6.44σ)
Mid-time O–C	-2041.6 min
Depth SNR	2.6
Residual scatter	71.28 %

Light curve

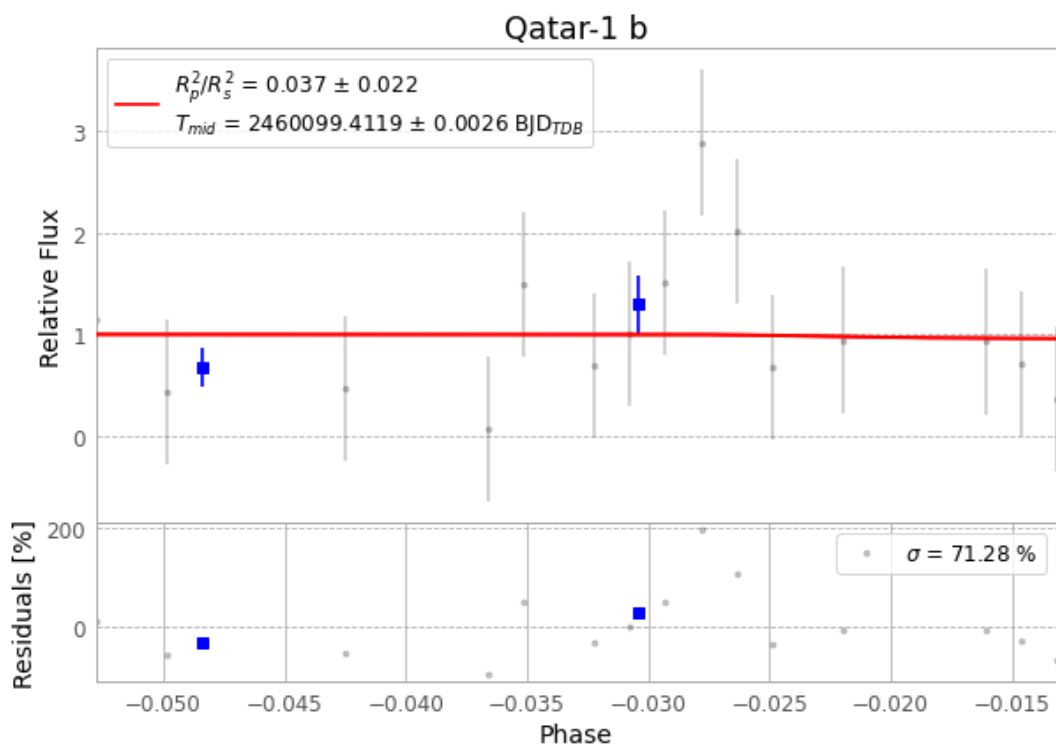


Figure 1 — FinalLightCurve_Qatar-1 b_2023-06-04.png (SHA-256 e82cf7ab15314f28...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [469, 455], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 0b5bbb16472dfe74...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit b365494

Data provenance

74 FITS frames (49 MB), Qatar-1230605060615.FITS → Qatar-1230605094515.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-230605.log (d14ed0fef9f3...), FinalLightCurve_Qatar-1 b_2023-06-04.png (e82cf7ab1531...), FinalLightCurve_Qatar-1 b_2023-06-04.csv (d84cfd8a3a6f...)

Compute resources

Wall time 115.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-25 14:32Z.